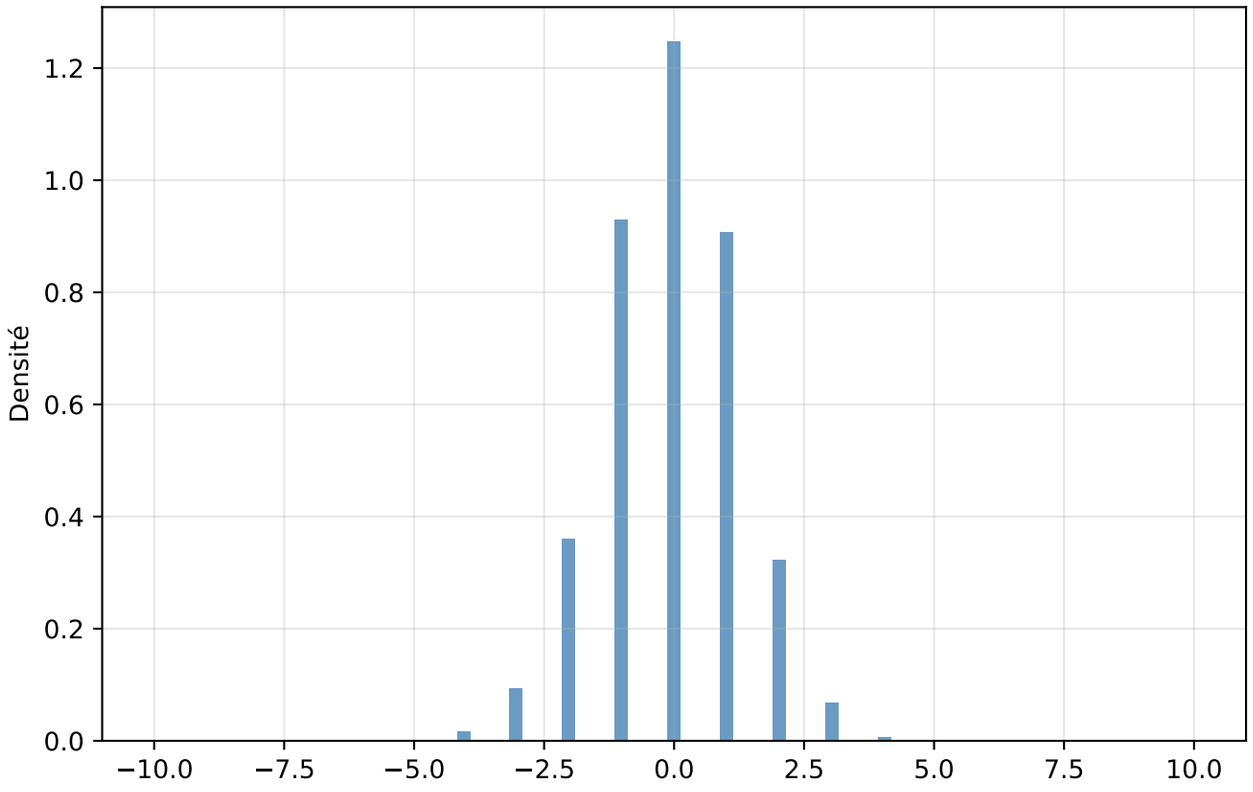
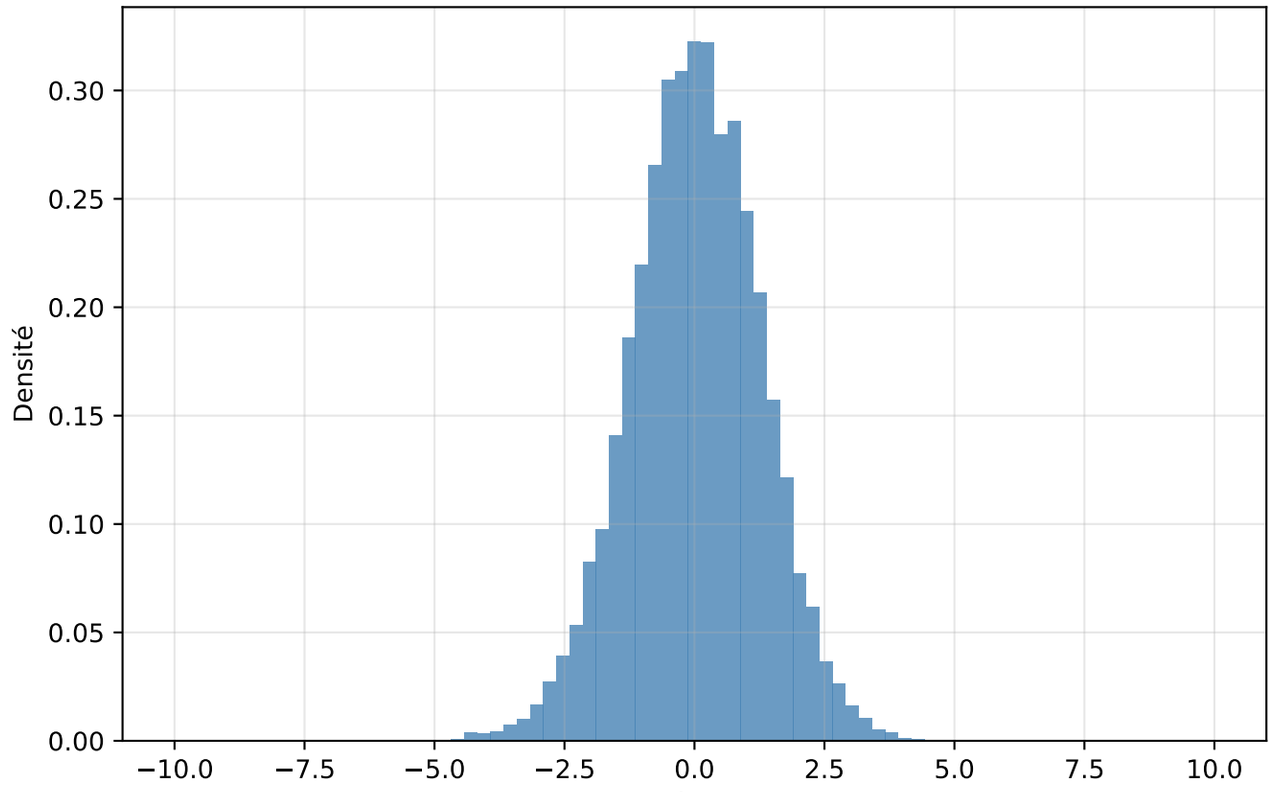
Distribution des résidus — argmax vs espérance softmax

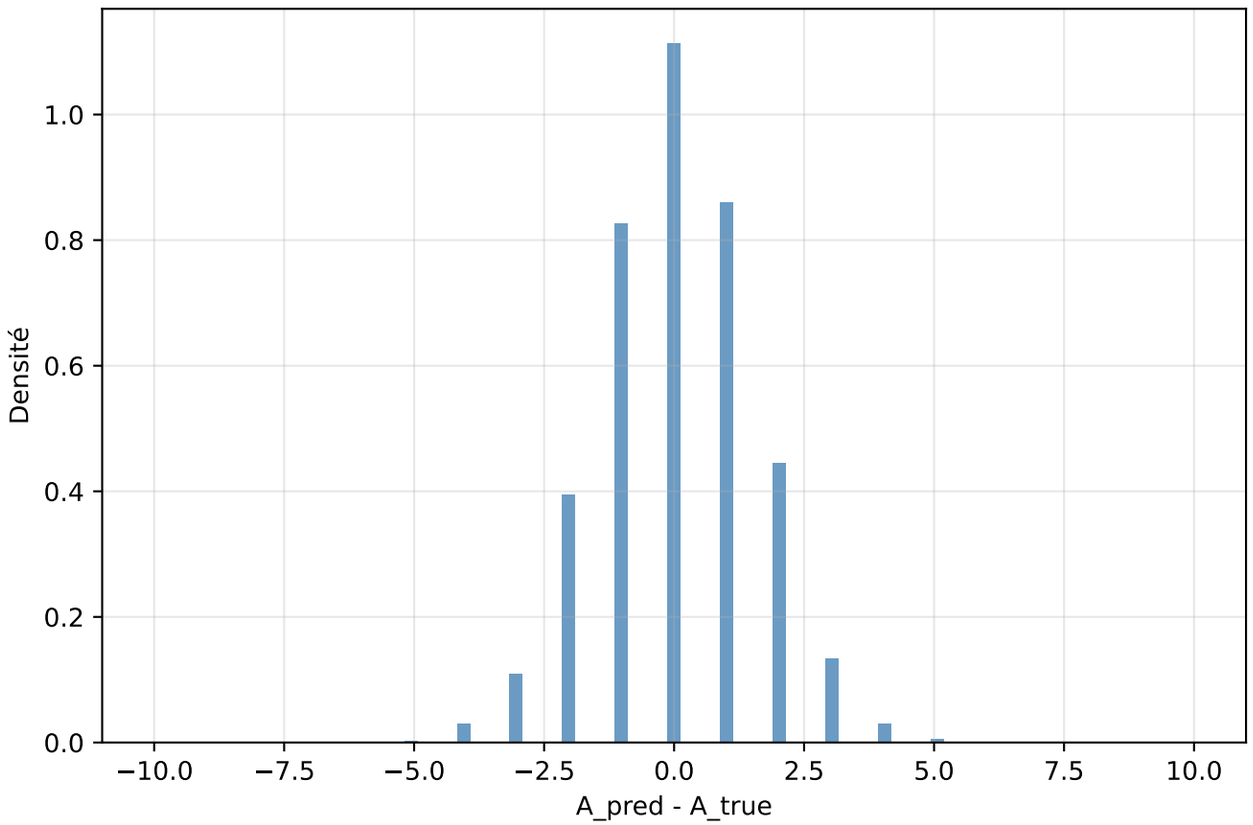
A_L argmax σ=1.288



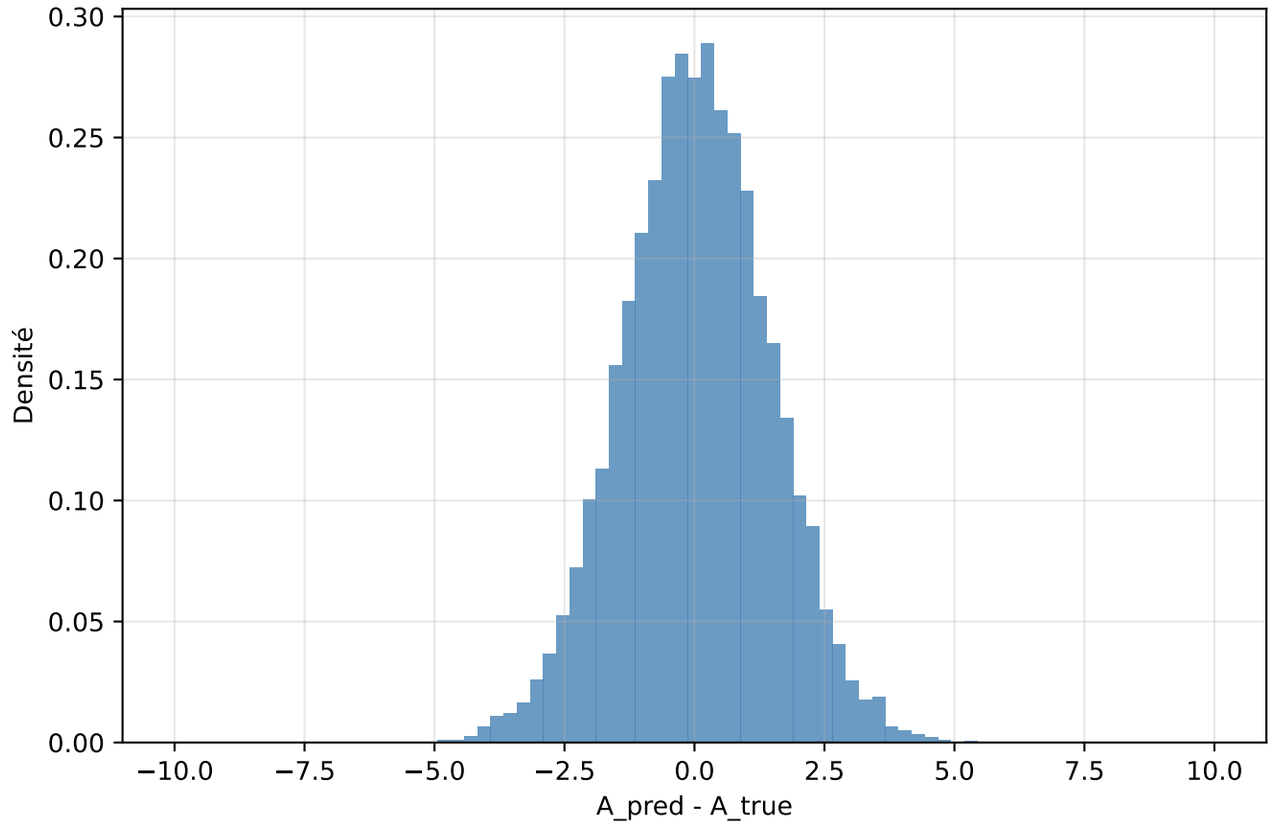
A_L espérance σ=1.242

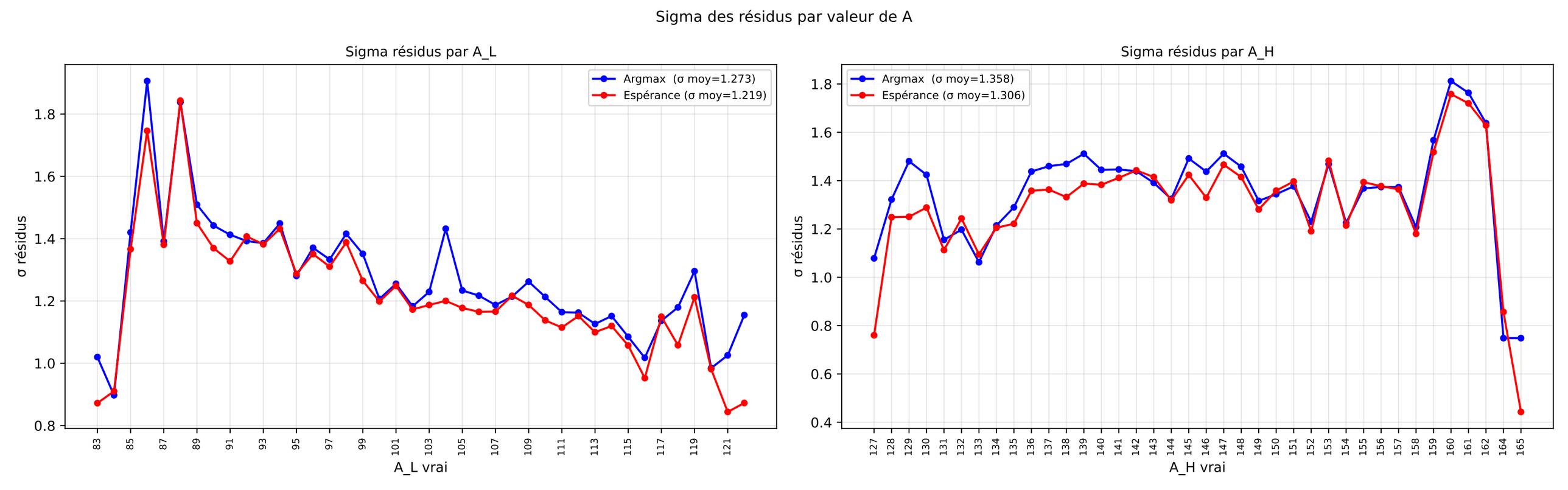


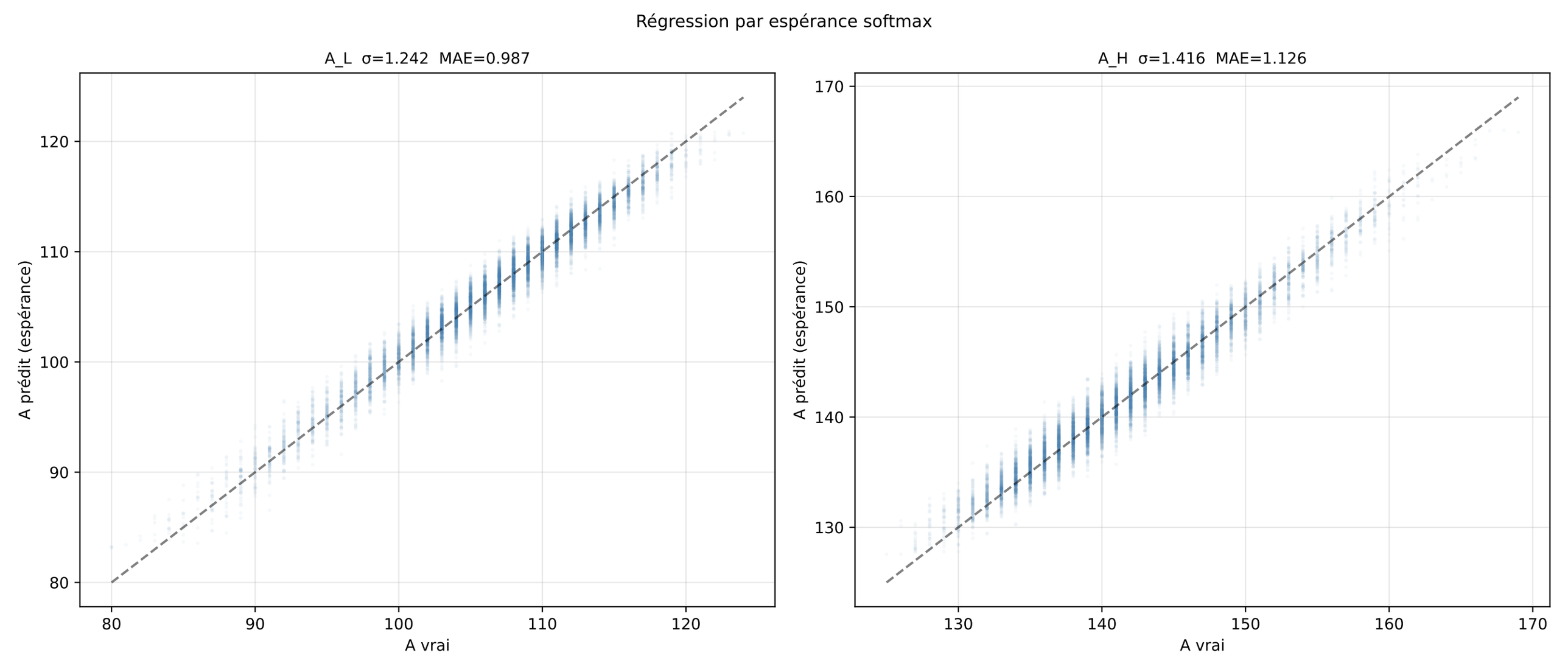
A_H argmax σ=1.457



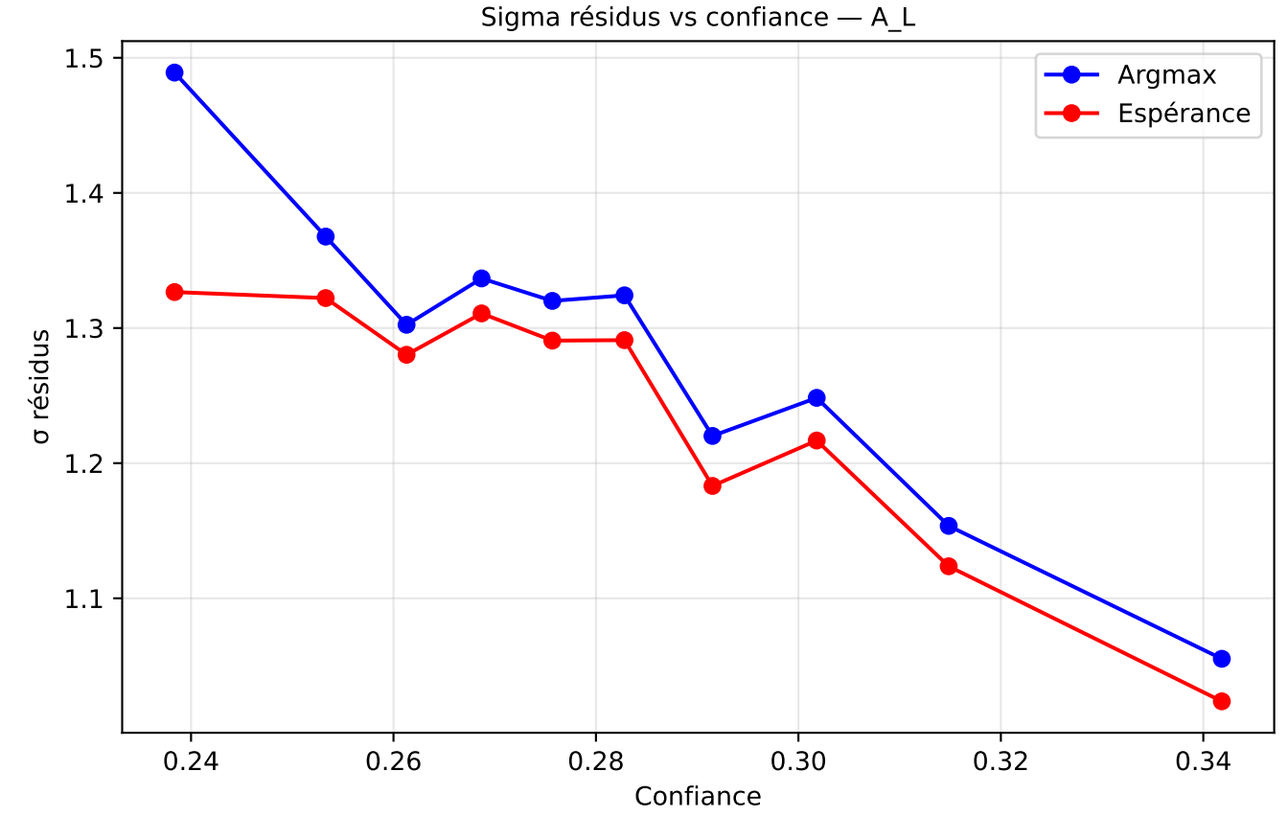
A_H espérance σ=1.416

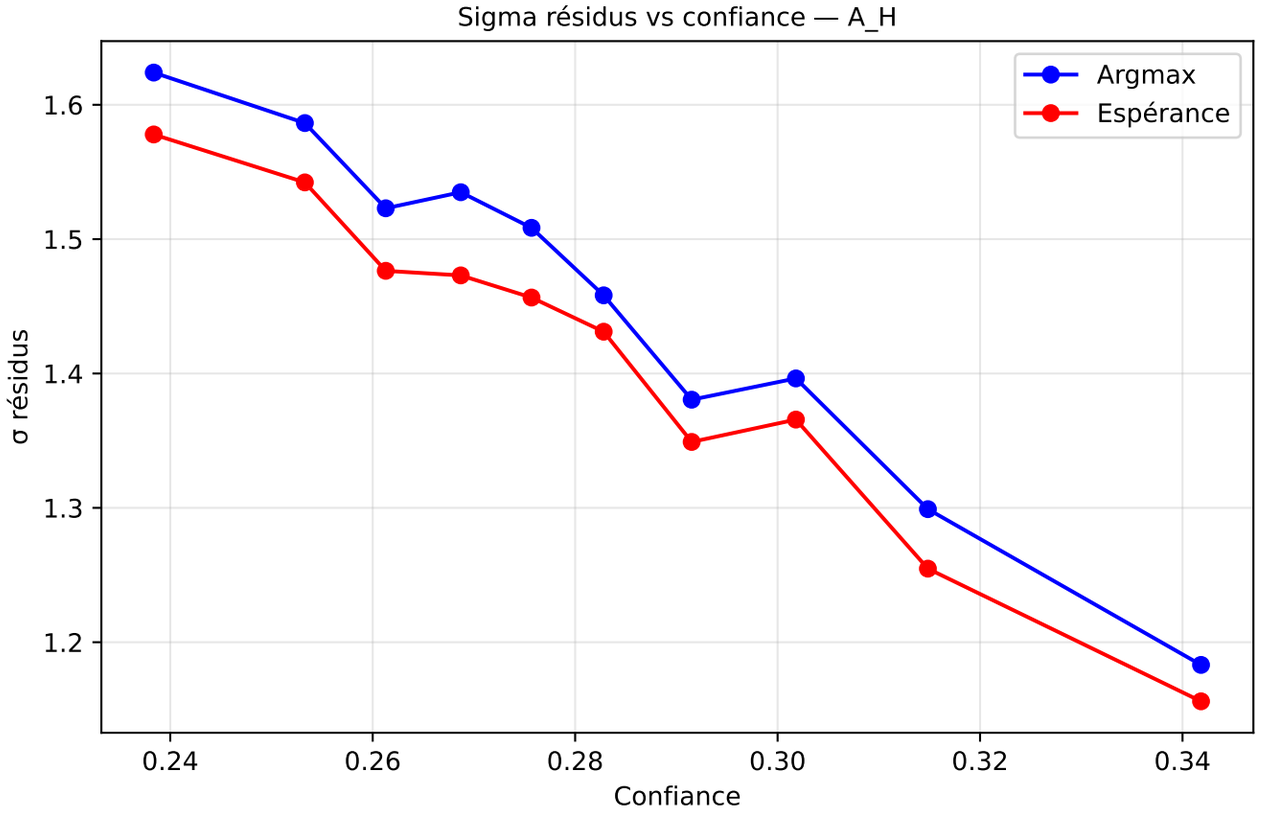


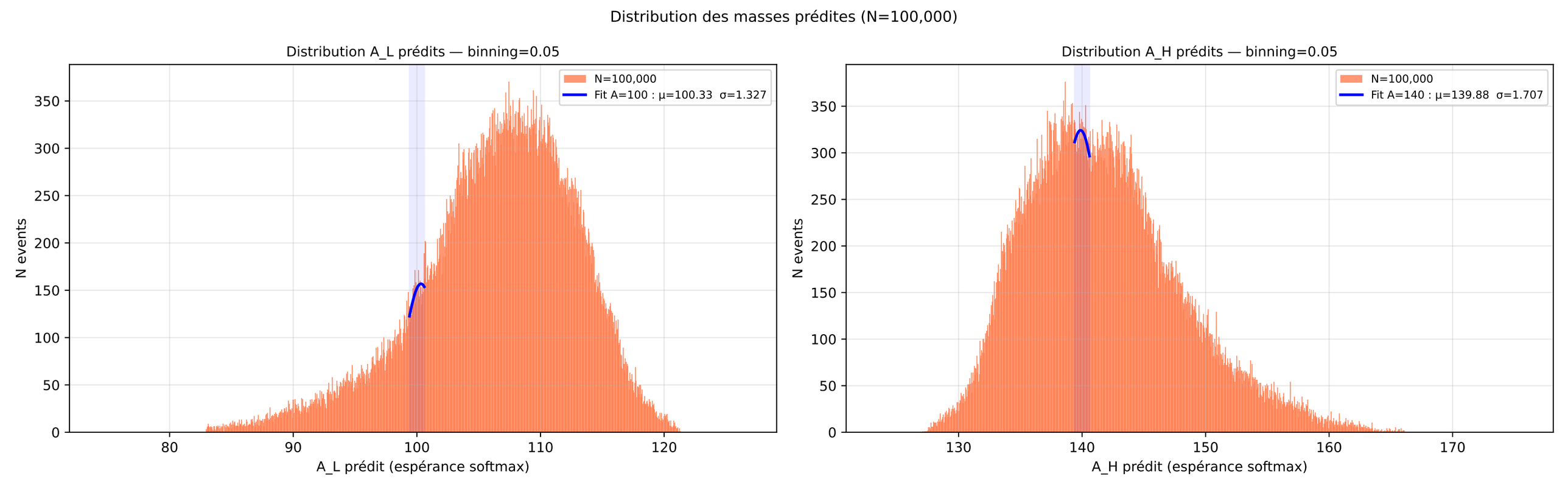

Sigma des résidus par valeur de A
Sigma résidus par A_L
Argmax (σ moy=1.273)
Espérance (σ moy=1.219)
σ résidus
A_L vrai
Sigma résidus par A_H
Argmax (σ moy=1.358)
Espérance (σ moy=1.306)
σ résidus
A_H vrai


Régression par espérance softmax
A_L σ=1.242 MAE=0.987
A_H σ=1.416 MAE=1.126
A prédit (espérance)
A vrai
80
90
100
110
120
130
140
150
160
170

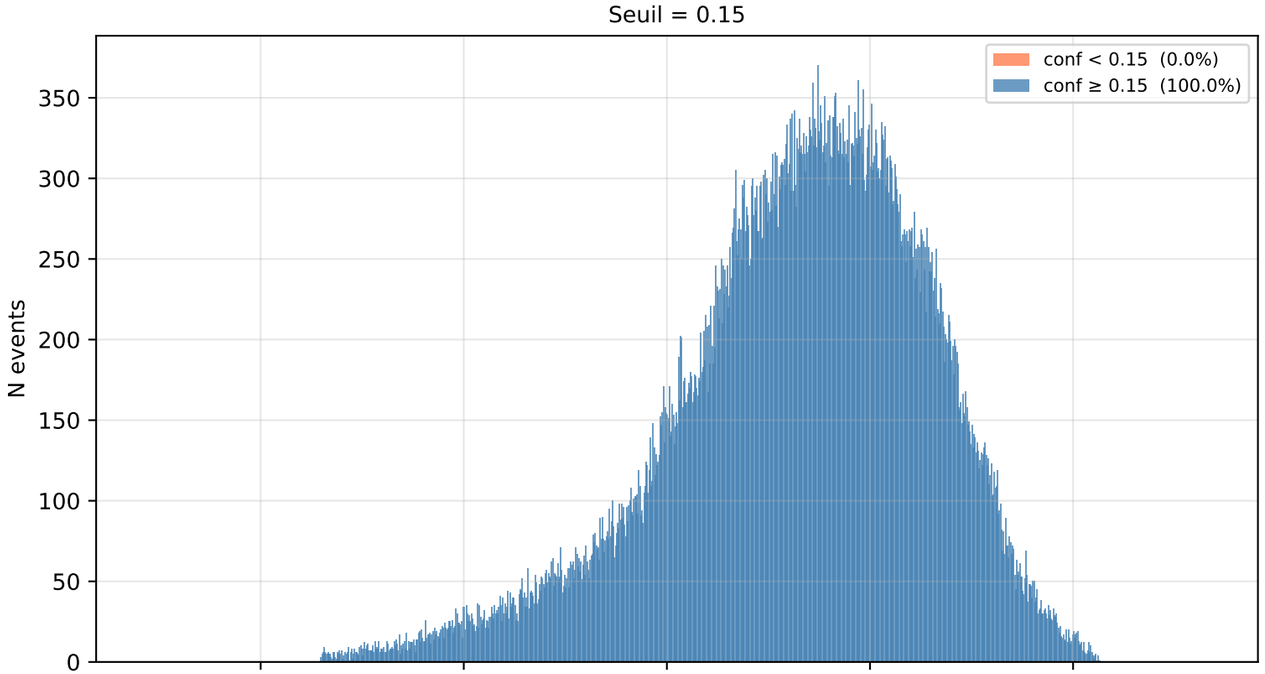
Les events incertains bénéficient-ils de la régression ?

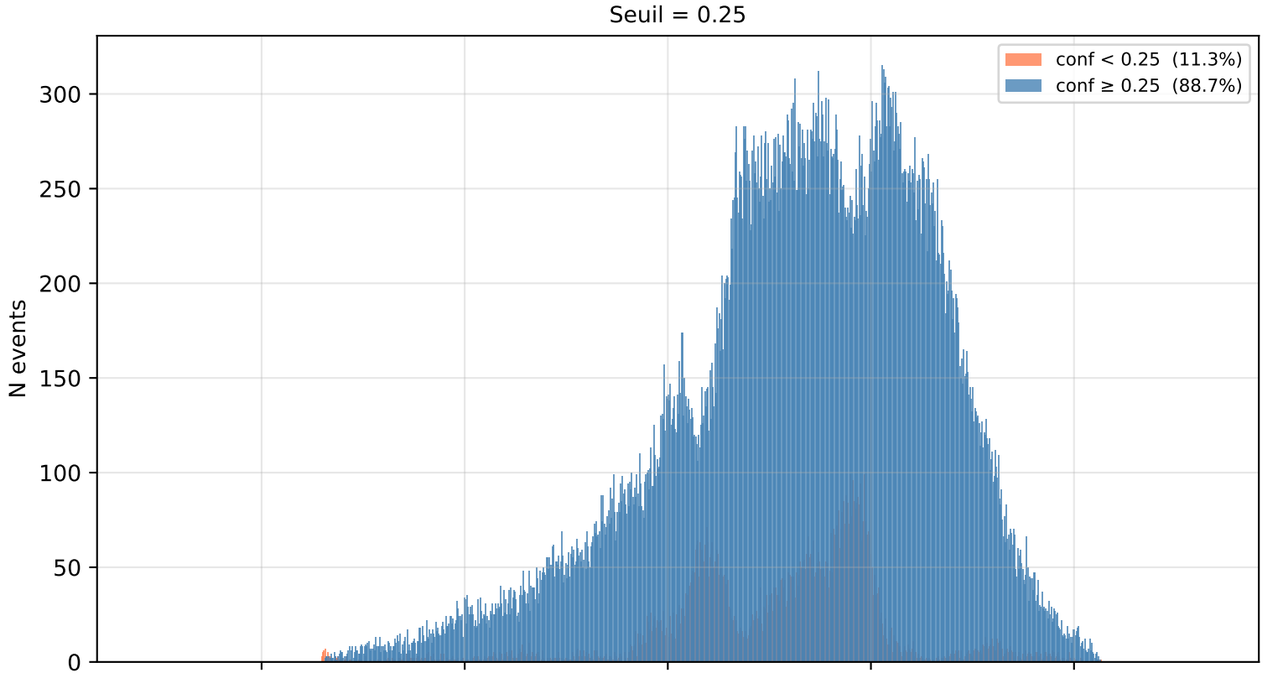


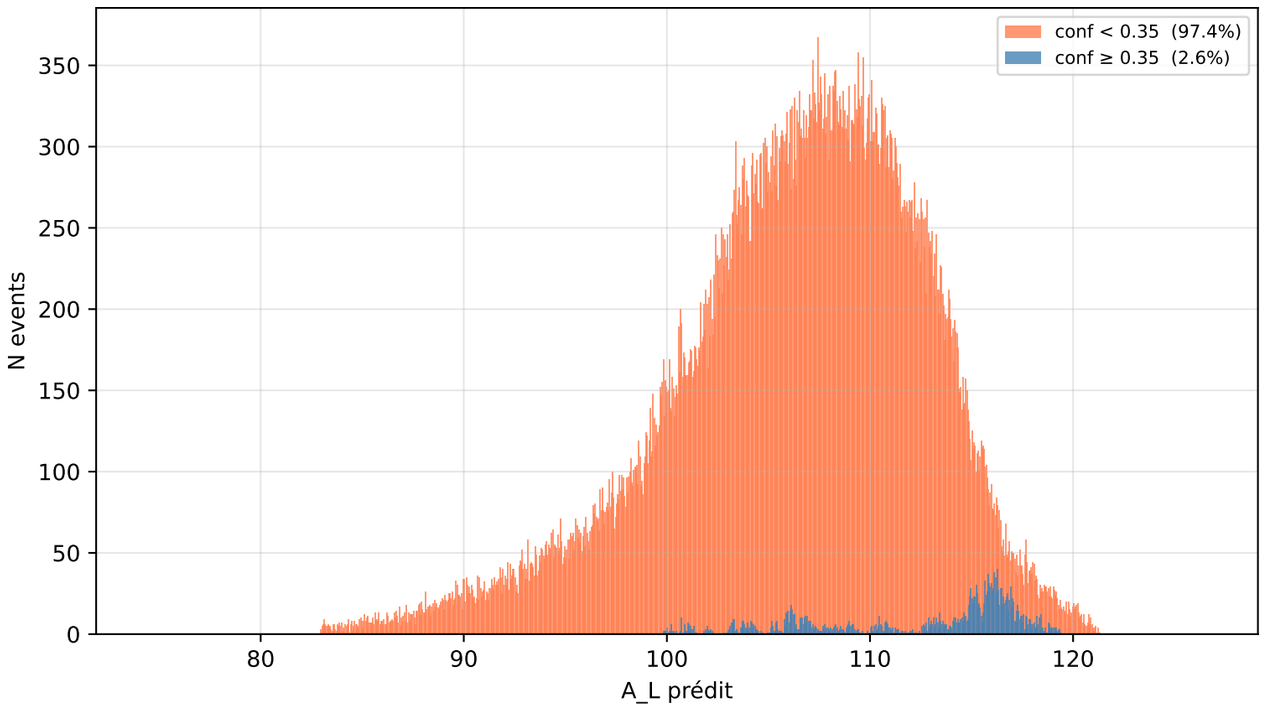


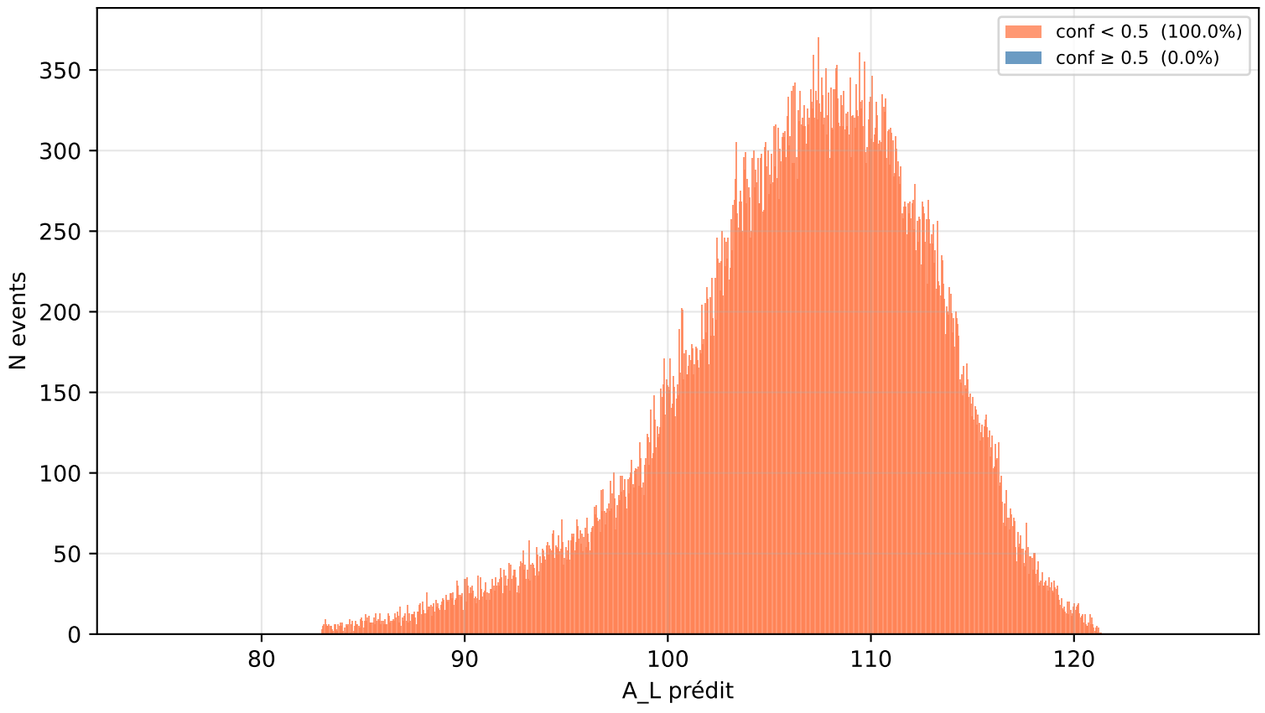

Distribution des masses prédites (N=100,000)
Distribution A_L prédits — binning=0.05
N=100,000
Fit A=100 : μ=100.33 σ=1.327
N events
A_L prédit (espérance softmax)
Distribution A_H prédits — binning=0.05
N=100,000
Fit A=140 : μ=139.88 σ=1.707
N events
A_H prédit (espérance softmax)

Distribution A_L prédits — séparation par seuil de confiance
Bleu : conf ≥ seuil | Orange : conf < seuil

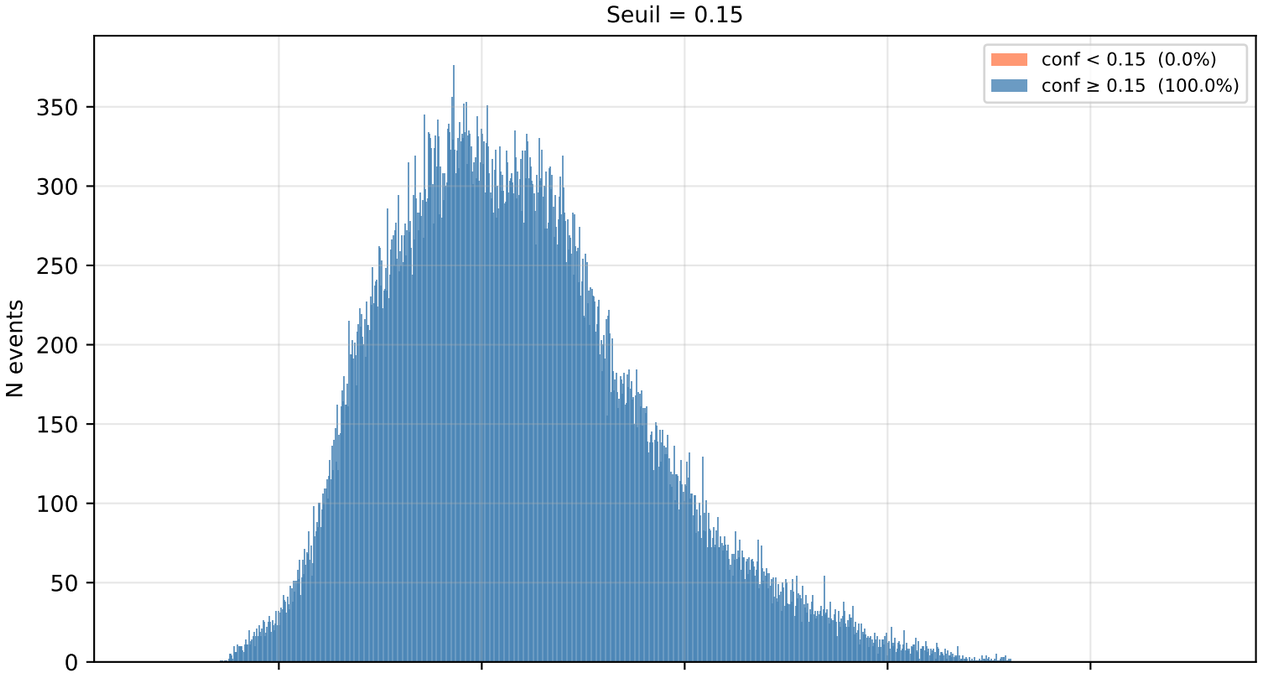


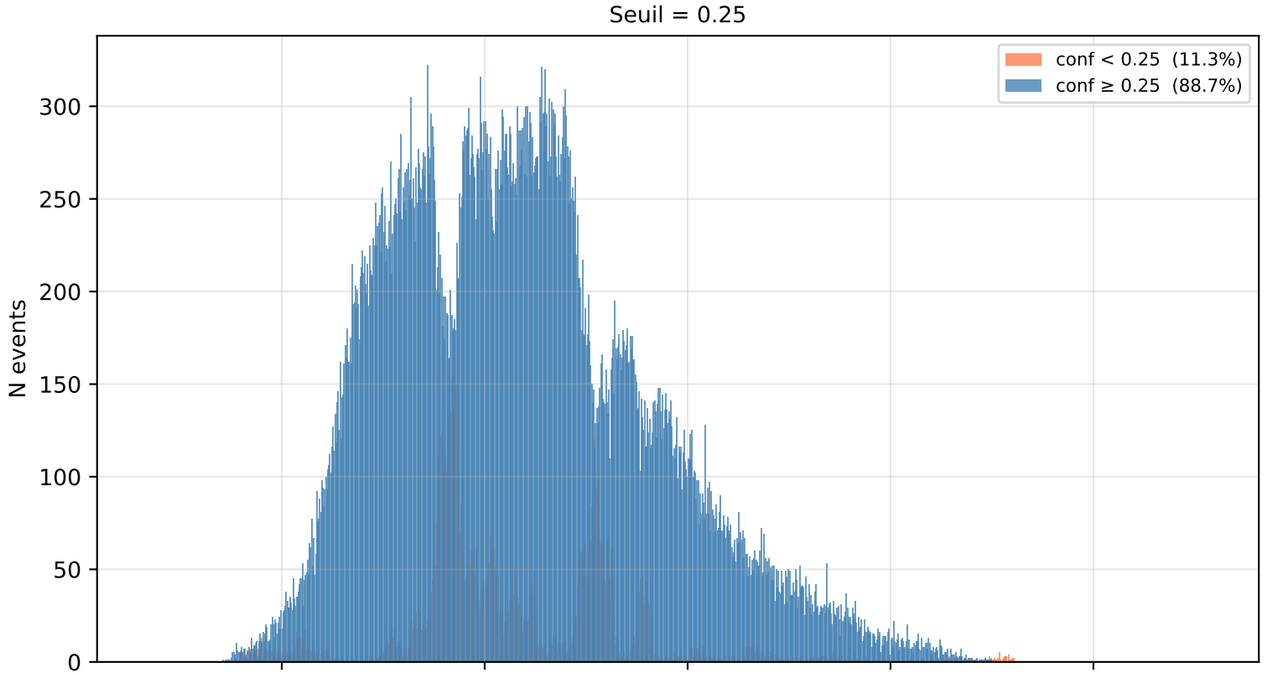


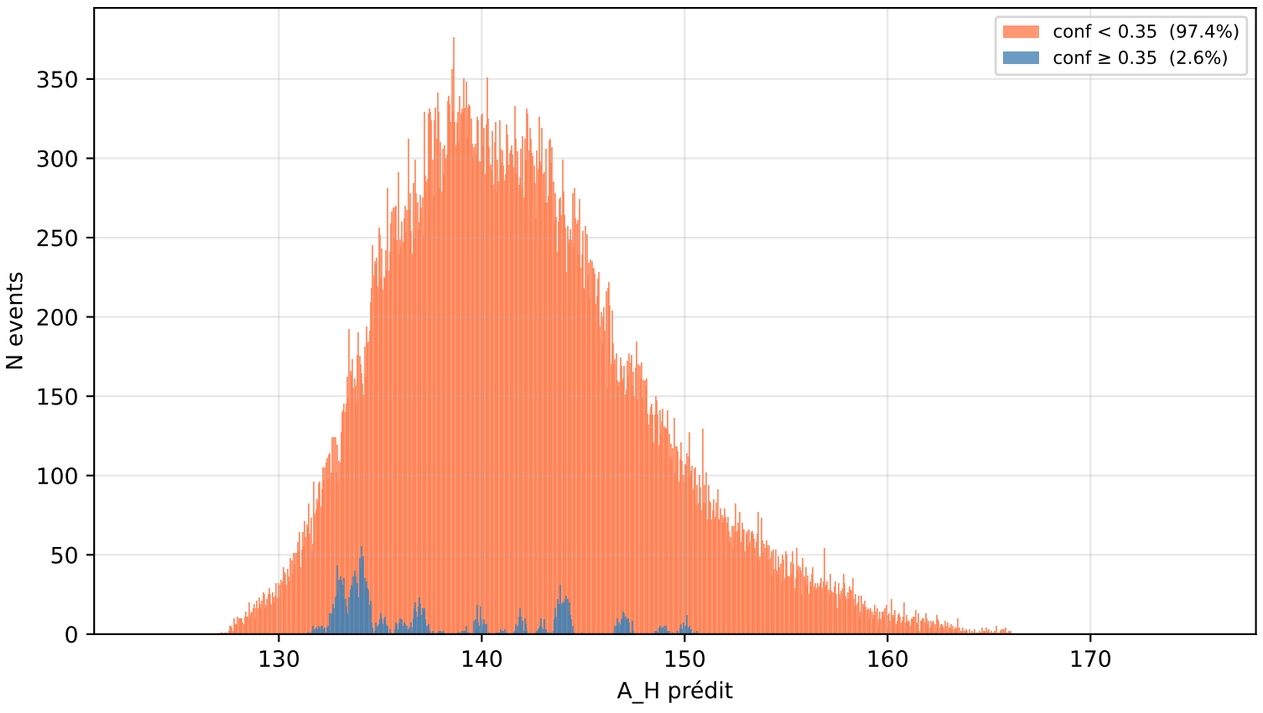


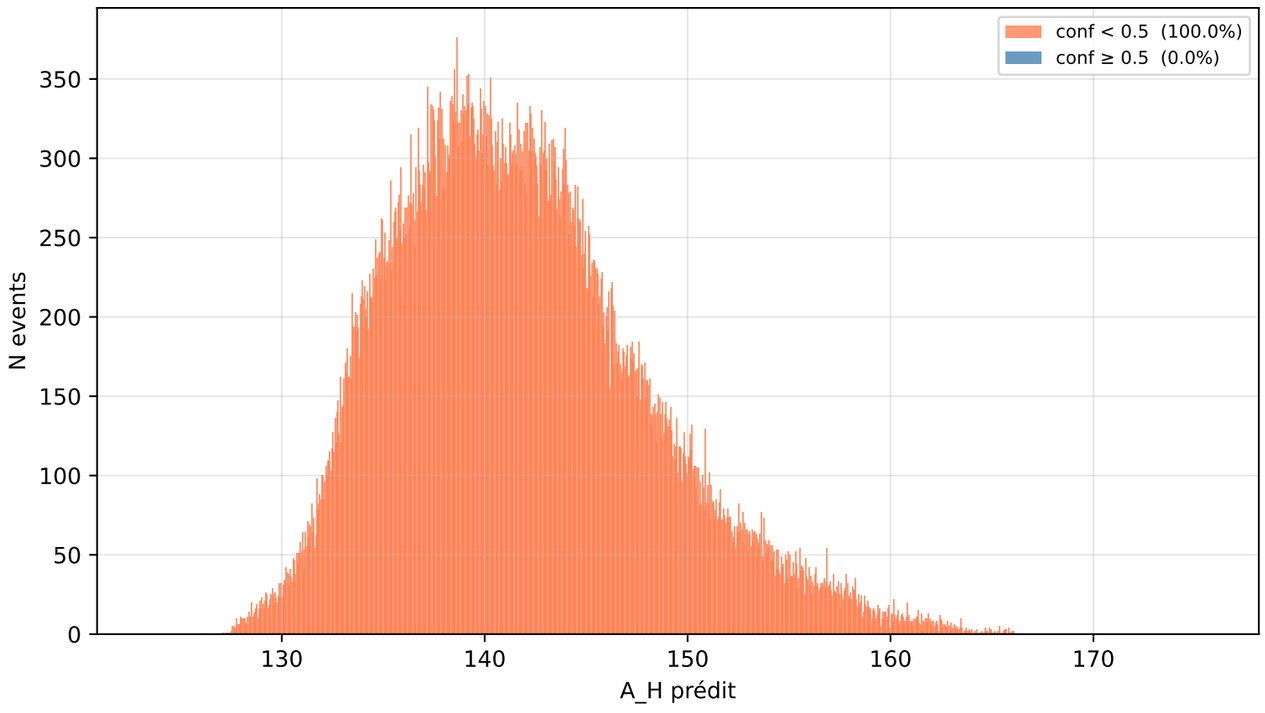


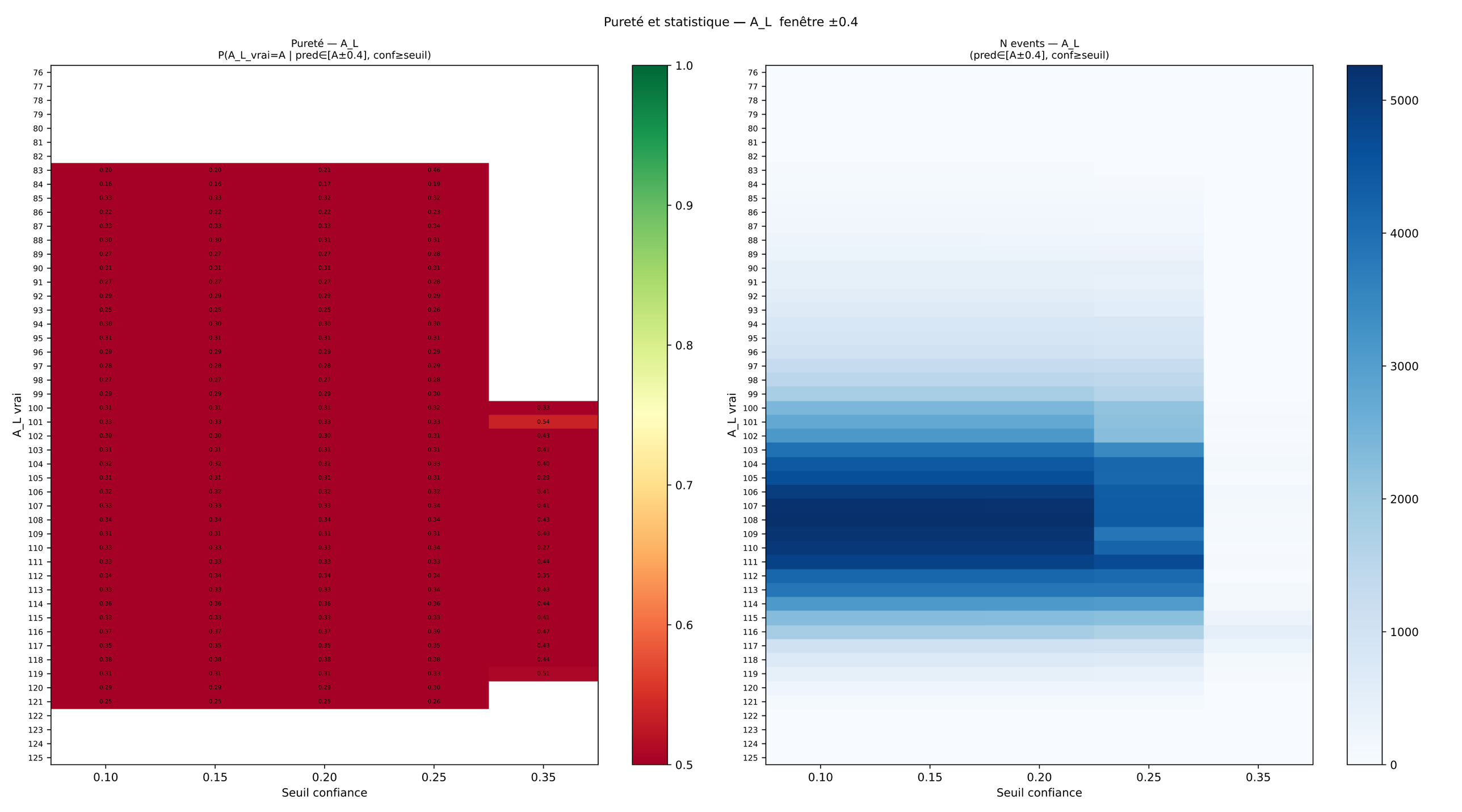
Distribution A_H prédits — séparation par seuil de confiance
Bleu : conf ≥ seuil | Orange : conf < seuil

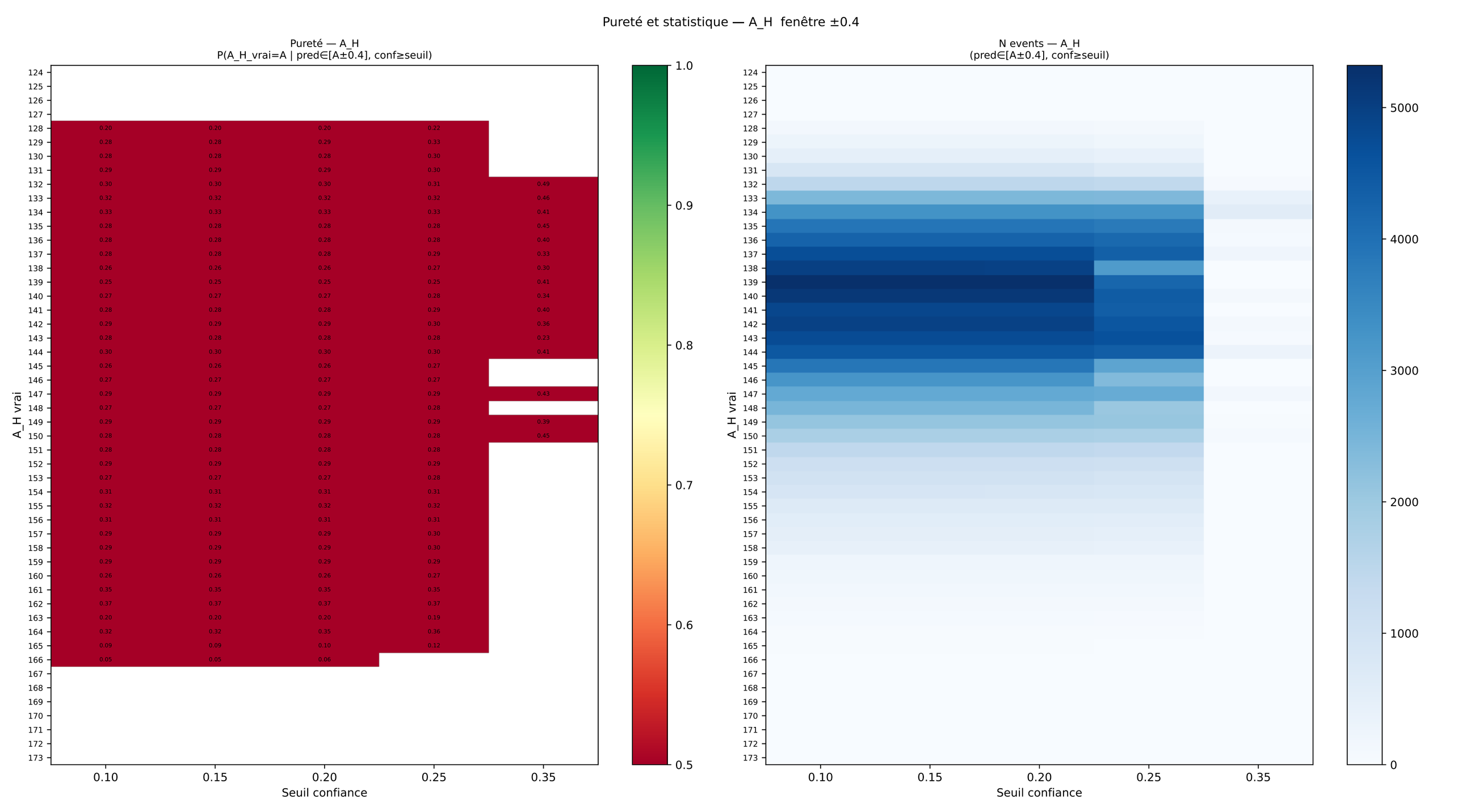


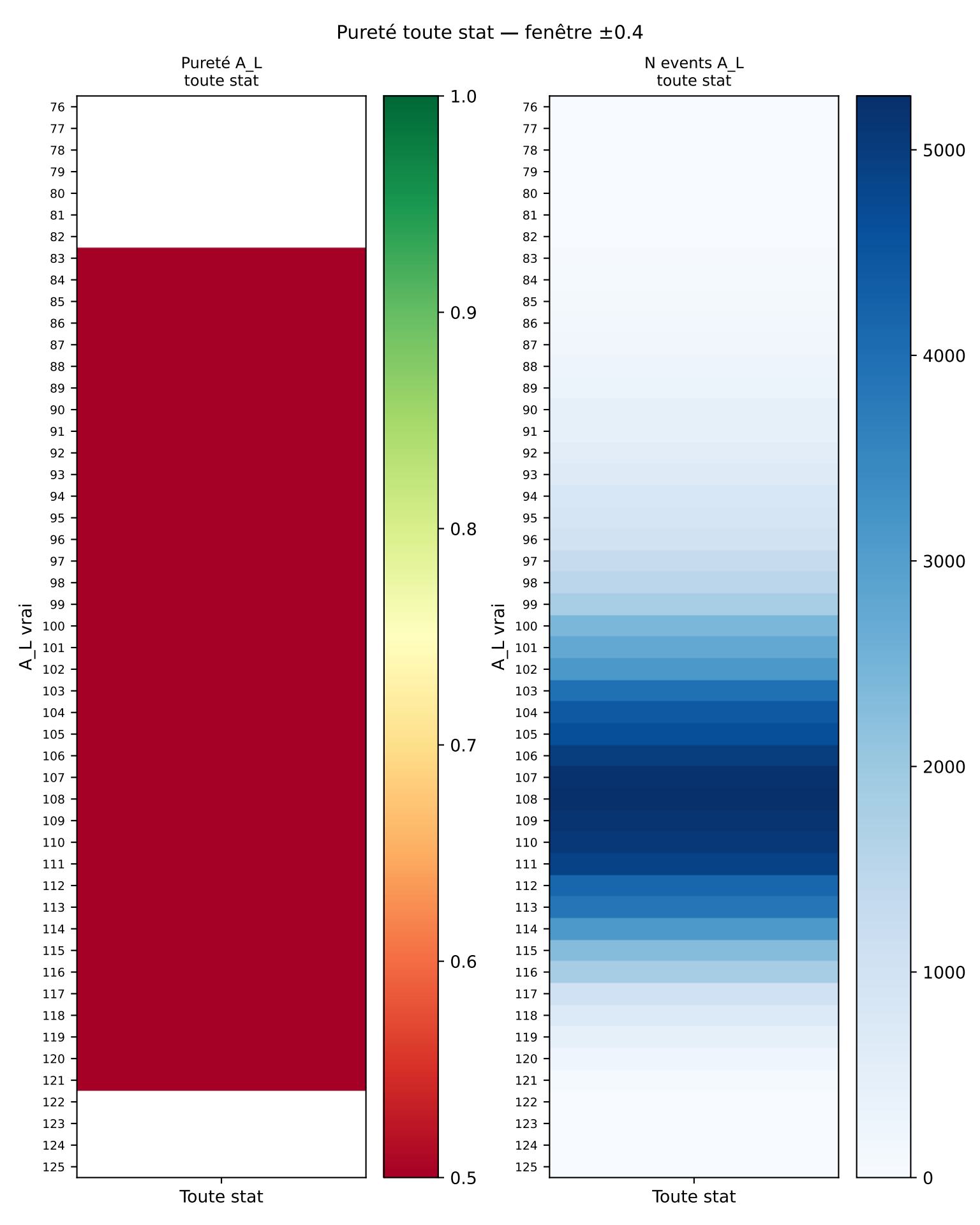







Pureté et statistique — A_L fenêtre ±0.4
Pureté — A_L
P(A_L_vrai=A | pred∈[A±0.4], conf≥seuil)
A_L vrai
Seuil confiance
0.10
0.15
0.20
0.25
0.35
1.0
0.9
0.8
0.7
0.6
0.5
N events — A_L
(pred∈[A±0.4], conf≥seuil)
5000
4000
3000
2000
1000
0


Pureté et statistique — A_H fenêtre ±0.4
Pureté — A_H
P(A_H_vrai=A | pred∈[A±0.4], conf≥seuil)
A_H vrai
Seuil confiance
N events — A_H
(pred∈[A±0.4], conf≥seuil)
A_H vrai
Seuil confiance


Pureté toute stat — fenêtre ±0.4
Pureté A_L
toute stat
N events A_L
toute stat
A_L vrai
Toute stat
1.0
0.9
0.8
0.7
0.6
0.5
5000
4000
3000
2000
1000
0